Improving Emergency Department Triage with a Hybrid Multi-Agent Verification Framework

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***Abstract*—Emergency Department (ED) triage plays a critical role in determining treatment priority and resource allocation for incoming patients. Recent advancements in Large Language Models (LLMs) have enabled the development of AI-based decision support systems capable of assisting healthcare professionals in triage classification tasks. This study investigates the effectiveness of three different AI architectures for Emergency Severity Index (ESI) prediction: a Single-Agent system, a traditional Multi-Agent Majority Voting system, and an Improved Hybrid Multi-Agent system. Experiments were conducted using 100 patient cases selected from the FEDMML-ED-Triage dataset. The Single-Agent architecture achieved 46% accuracy, while the Multi-Agent Majority Voting architecture achieved only 31% accuracy, performing below expectations. The proposed Improved Hybrid Multi-Agent system, incorporating a verification layer, clinical safety rules, and human-review escalation, achieved the highest performance with 65% accuracy. The findings indicate that verification and safety-oriented coordination mechanisms are essential components for reliable multi-agent clinical decision support systems.**

***Index Terms*—Emergency Department Triage, Large Language Models, Multi-Agent Systems, Clinical Decision Support, Emergency Severity Index, Artificial Intelligence**

I. Introduction

Emergency Departments (EDs) face continuous challenges in prioritizing patients with varying levels of urgency while operating under limited resources and time constraints. Triage systems are designed to ensure that critically ill patients receive immediate medical attention while lower-acuity cases are managed efficiently [1].

One of the most widely used triage frameworks is the Emergency Severity Index (ESI), a five-level categorization system ranging from ESI-1 (most critical) to ESI-5 (least urgent). Accurate ESI classification is essential because under-triage may delay treatment for life-threatening conditions, whereas over-triage leads to inefficient resource utilization and increased waiting times [7].

Recent advances in Artificial Intelligence (AI) and Large Language Models (LLMs) have introduced new opportunities for automated clinical decision support [8]. LLM-based agents can process patient records, interpret clinical notes, analyze vital signs, and generate triage recommendations. However, the effectiveness of different agent architectures in emergency triage remains an open research question.

Traditional single-agent architectures rely on a single model to analyze all available patient information and generate a final decision. While such systems are simple and computationally efficient, they may overlook specialized medical perspectives. Multi-agent systems attempt to address this limitation by introducing domain-specific agents that independently evaluate patient cases and collaboratively determine a final outcome [3]. Despite their theoretical advantages, naive multi-agent architectures may suffer from specialist disagreement, information fragmentation, and coordination challenges. Therefore, simply increasing the number of agents does not necessarily guarantee improved decision quality.

This paper investigates three architectures for AI-assisted emergency triage: (1) a Single-Agent baseline, (2) a Multi-Agent Majority Voting architecture, and (3) an Improved Hybrid Multi-Agent architecture. The objective is to evaluate whether specialist collaboration, verification mechanisms, and human-review escalation can improve ESI classification performance.

The primary contributions of this paper are:

1) Design of a multi-agent framework for emergency department ESI triage incorporating specialist agents, a verification layer, and clinical safety rules.
2) Experimental comparison of Single-Agent, Multi-Agent, and Improved Hybrid Multi-Agent architectures on a real-world synthetic ED triage dataset.
3) Analysis of the strengths, limitations, and failure modes of multi-agent clinical decision support systems, together with proposed mitigation strategies.
4) Evidence that verification and human-escalation mechanisms are essential components of reliable agentic clinical AI, particularly for high-acuity patient safety.

II. Related Work

A. LLM-Based Clinical Decision Support

The application of large language models to clinical decision support has grown rapidly. Topol [8] provided an early vision of AI transforming healthcare, arguing that deep learning systems could augment clinical judgment across a wide range of tasks. The release of powerful general-purpose models such as GPT-4 [2] further demonstrated strong performance on medical licensing examinations and clinical reasoning benchmarks, indicating the potential of LLMs as clinical assistants.

However, general-purpose LLMs are not inherently designed for high-stakes medical decisions. Domain-specific

adaptations, structured prompting strategies, and verification mechanisms have emerged as essential components to ensure reliable outputs in clinical settings [9]. Sutton et al. [9] reviewed the landscape of clinical decision support systems, emphasizing the need for transparency, explainability, and integration with clinical workflows.

B. Multi-Agent and Agentic AI Systems

Multi-agent systems (MAS) have been extensively studied in artificial intelligence research [3]. In MAS, multiple autonomous agents interact through defined communication protocols, share information, and coordinate to achieve shared objectives. Key advantages include parallelism, redundancy, and the potential to combine specialized capabilities across agents.

Wang et al. [11] provided a comprehensive survey of LLM-based autonomous agent architectures, highlighting the importance of memory management, tool use, planning, and coordination mechanisms. Their taxonomy distinguishes between single-agent and multi-agent paradigms and identifies coordination as the central challenge in multi-agent LLM systems.

Hong et al. [10] proposed MetaGPT, a multi-agent collaborative framework in which LLM-based agents adopt specialized roles (engineer, product manager, architect) and coordinate through structured communication channels. Their work demonstrated that role specialization within LLM-based multi-agent systems can improve performance on complex tasks when accompanied by structured communication protocols. This motivates the investigation of coordination mechanisms in our clinical triage setting.

C. Agentic AI for Medical Reasoning

Li et al. [4] introduced MedAgents, a framework in which multiple LLM-based specialist agents collaborate on medical reasoning tasks through multi-round discussion protocols. Their experiments showed that properly coordinated multi-agent collaboration can outperform single-agent approaches on medical question-answering benchmarks. This work directly motivates our investigation of multi-agent designs for emergency triage and highlights the importance of coordination mechanisms beyond simple majority voting.

Du et al. [12] demonstrated at ICML 2024 that structured *multiagent debate* – in which independently reasoning language models iteratively critique each other's responses and refine their positions – substantially improves factual accuracy and reasoning quality compared to single-model inference or naive aggregation. Their findings directly inform the improvement opportunities explored in Section VII of this paper, where debate-style inter-agent communication is proposed as a mitigation for the specialist disagreement failures identified in our system.

D. Emergency Triage and AI

Triage reliability and validity have been studied within clinical informatics for decades [7]. The ESI framework has been validated as a reliable five-level scale with strong inter-rater agreement among trained clinical staff. Automated triage tools have been explored using rule-based systems, machine learning classifiers, and, more recently, deep learning models trained on structured clinical data. Sutton et al. [9] concluded that successful deployment of clinical decision support requires transparency, explainability, and tight integration with existing workflows – requirements that are especially stringent in triage.

Our work extends these efforts by applying a multi-agent LLM-based architecture specifically to ESI prediction. Unlike prior machine learning approaches requiring large labeled training sets and offline model training, our system operates entirely through prompted inference, making it applicable even in settings with limited historical data.

E. Summary of Research Gaps

Three key gaps motivate this work. First, most multi-agent medical AI studies evaluate on multiple-choice benchmarks rather than on structured triage classification that requires integrating heterogeneous clinical data modalities. Second, the failure modes of naive multi-agent majority voting in clinical contexts are poorly characterized: existing work tends to report only positive results. Third, the role of hybrid symbolic–neural verification layers in improving agentic clinical AI reliability has not been systematically studied. This paper directly addresses all three gaps through a controlled comparative evaluation.

III. System Design

A. Agent Roles

The proposed Improved Hybrid Multi-Agent system incorporates five distinct agent roles, each responsible for a specific function in the triage pipeline.

Single-Agent Assessor: A generalist emergency physician agent that receives the complete patient record and produces an initial ESI assessment. This agent serves as a first-pass reference and a consistency anchor for downstream specialist agents. *Input:* full patient record. *Output:* preliminary ESI level (1–5) with supporting reasoning.

Cardiologist Agent: Specializes in cardiac risk factors, analyzing blood pressure, heart rate, troponin levels, BNP, and cardiac-related chief complaints. *Input:* full patient record. *Output:* ESI recommendation and cardiac reasoning.

Trauma Specialist Agent: Focuses on trauma-related indicators including injury mechanism, respiratory rate, oxygen saturation (SpO2), and lactate levels. *Input:* full patient record. *Output:* ESI recommendation and trauma-specific reasoning.

General Practitioner Agent: Evaluates the overall clinical picture including chief complaint, vital signs, and laboratory findings from a general medicine perspective. *Input:* full patient record. *Output:* ESI recommendation and general clinical reasoning.

Verification Agent: Reviews all specialist recommendations alongside critical clinical indicators. Applies predefined clinical safety rules, identifies specialist disagreement, and

produces the final ESI decision or flags the case for human escalation. *Input:* all specialist outputs and patient vital signs. *Output:* final ESI level or human-escalation flag.

B. Workflow

The workflow of the Improved Hybrid Multi-Agent system proceeds through four sequential stages. Fig. 1 illustrates the Multi-Agent Majority Voting pipeline, while Fig. 2 illustrates the full Improved Hybrid Multi-Agent pipeline.

Stage 1 – Initial Assessment: The patient record is forwarded to the Single-Agent Assessor, which produces a preliminary ESI recommendation.

Stage 2 – Specialist Review: The same patient record is forwarded to the Cardiologist, Trauma Specialist, and General Practitioner agents, which operate in parallel. Each agent independently generates an ESI recommendation and clinical reasoning.

Stage 3 – Verification and Safety Rules: The Verification Agent collects all specialist outputs and the initial assessment. It applies a rule-based clinical safety check examining: SpO2 $< 90\%$, systolic blood pressure < 90 mmHg or > 180 mmHg, heart rate > 130 bpm, respiratory rate > 30/min, troponin elevation, lactate > 4 mmol/L, and critical complaint keywords (chest pain, stroke, sepsis, respiratory failure). If any critical threshold is violated, a minimum ESI level of 2 is enforced regardless of specialist votes.

Stage 4 – Human Escalation: If three or more specialists disagree substantially (spread ≥ 2 ESI levels), the case is flagged for mandatory human review. Otherwise, the Verification Agent produces a final ESI decision.

C. Design Rationale

The central design rationale stems from an empirical observation: naive majority voting across specialist agents does not reliably resolve disagreements and can produce worse outcomes than a single generalist agent. This observation motivated three key design decisions.

First, a single-agent initial assessment was retained as a consistency anchor. The initial assessment gives the verification layer a holistic patient evaluation, helping to counterbalance over-specialization by individual agents.

Second, clinically grounded safety rules were embedded directly into the verification layer. Rather than relying solely on LLM reasoning for critical decisions, hard-coded thresholds for vital sign abnormalities and critical complaint keywords enforce minimum safety guarantees. This hybrid symbolic–neural design is consistent with recommendations in clinical AI literature [9].

Third, human escalation was introduced as a fallback mechanism. When agents disagree substantially, the system acknowledges the limits of automated reasoning and defers to human expertise, reducing the risk of systematic errors in ambiguous cases [3].

IV. Implementation

A. Development Environment

The system was implemented in Python 3.11. All experiments were conducted on a standard laptop without GPU acceleration, as language model inference was performed remotely through a cloud API.

B. Frameworks and Libraries

- **Groq API** [6]: Used for all LLM inference. Groq's high-throughput, low-latency infrastructure supported the `llama3-70b-8192` model, selected for its balance of reasoning capability and inference speed. Groq's higher requests-per-minute limits compared to standard providers were critical for processing 100 cases with multiple sequential agent calls per case.
- **Hugging Face Datasets**: Used to load and sample the FEDMML-ED-Triage dataset [5].
- **Pandas**: Used for data manipulation, result storage, and tabular reporting.
- **Scikit-learn**: Used for computing precision, recall, F1-score, and accuracy metrics.

C. Prompt Engineering

Each agent was instantiated with a distinct system prompt encoding its specialist role. Prompts were structured to instruct agents to: (1) identify the three most clinically significant findings relevant to their specialty; (2) assign an ESI level from 1 to 5 with explicit step-by-step chain-of-thought reasoning; and (3) format output as a structured JSON object for downstream parsing. An example JSON schema for a specialist agent is:

```
{
  specialist: Cardiologist,
  critical_findings: [...],
  esi_level: 2,
  reasoning: ...
}
```

For the Verification Agent, the system prompt additionally included the full clinical safety rule checklist, explicitly instructing the agent to cross-reference specialist votes against the predefined thresholds before issuing a final decision. Zero-shot chain-of-thought prompting was used throughout: agents were explicitly instructed to reason through findings before stating a conclusion, which empirically reduced hallucinated clinical justifications compared to direct-answer prompting. Role-specific prompts were essential; preliminary experiments using a single generic prompt for all agents produced noticeably lower-quality specialist reasoning.

D. Agent Orchestration

A central Python orchestrator managed the agent pipeline sequentially for each patient case:

1) Format the patient record into a standardized clinical note string.
2) Invoke the Single-Agent Assessor.

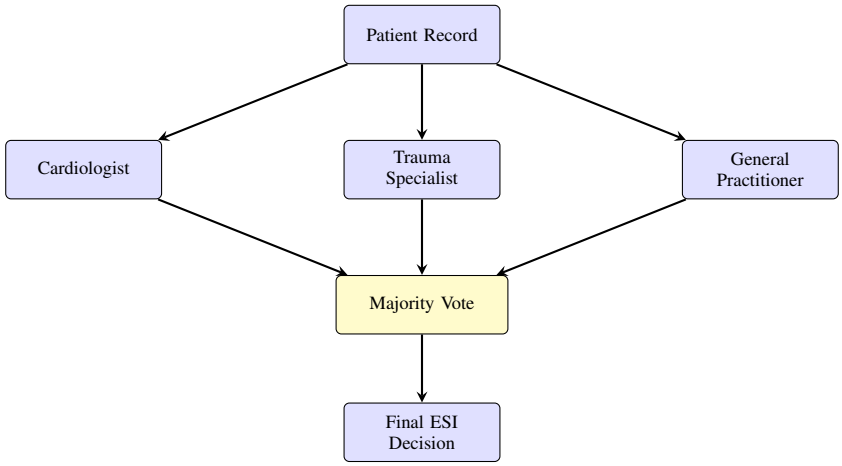


Fig. 1. Multi-Agent Majority Voting architecture. Three specialist agents independently evaluate the patient record; the final ESI decision is determined by majority vote.

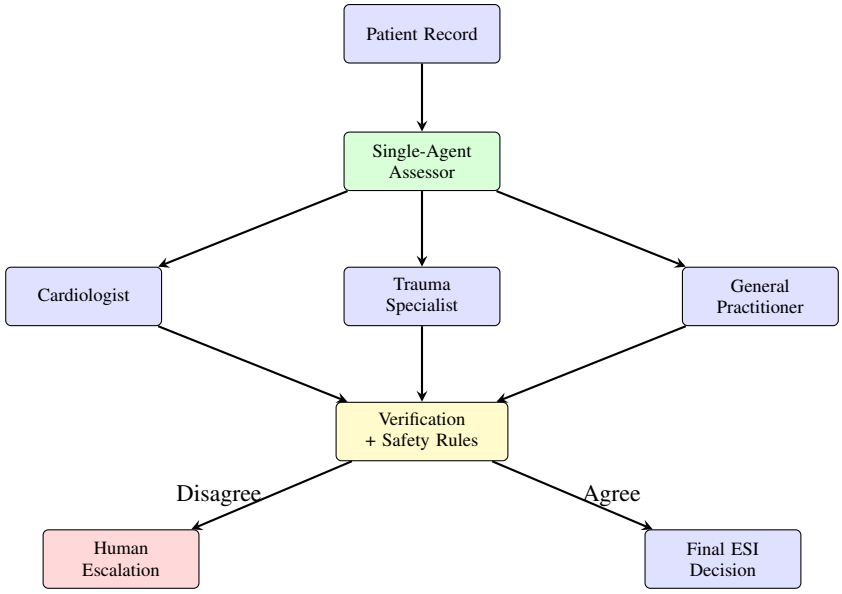


Fig. 2. Improved Hybrid Multi-Agent architecture. The patient record passes through an initial single-agent assessment, a specialist layer, a verification and clinical-rules layer, and a human-escalation layer before producing the final ESI decision.

3) Invoke the three specialist agents in parallel using Python's `concurrent.futures` module to reduce total latency.
4) Collect all specialist outputs and forward them to the Verification Agent together with the patient's structured vital signs.
5) Apply rule-based safety checks on structured vital sign data against predefined clinical thresholds.
6) If the disagreement threshold is exceeded, flag the case for human review; otherwise record the final ESI decision.

E. Memory Mechanisms

Each agent call was stateless: no memory was maintained across patient cases. Each invocation received the full patient record and any upstream agent outputs relevant to that agent's role. This design ensured isolation between cases and prevented cross-case information leakage, which is a critical safety requirement in clinical applications.

F. Verification Components

The verification layer comprised two components: (1) a rule-based safety checker implemented in Python that evaluated structured vital sign data against predefined clinical thresholds, and (2) an LLM-based synthesis call that integrated specialist votes with safety rule outputs to produce a coherent final recommendation. The rule-based component always overrode the LLM component in safety-critical situations, ensuring that hard clinical boundaries were respected regardless of model outputs.

V. Experimental Setup

A. Research Questions

The following research questions guided the experimental design:

RQ1: Does a multi-agent specialist system outperform a single-agent generalist system for ESI triage classification?

RQ2: Does adding a verification layer and clinical safety rules to a multi-agent system improve overall classification performance beyond both single-agent and naive multi-agent baselines?

RQ3: Are verification and safety mechanisms particularly beneficial for high-acuity (ESI-1 and ESI-2) patients, where under-triage has the most severe clinical consequences?

B. Baselines

Two baseline systems were evaluated:

Baseline 1 – Single-Agent: A single LLM acting as a generalist emergency physician, receiving the complete patient record and directly assigning an ESI level. This is the simplest possible architecture and serves as the primary performance reference.

Baseline 2 – Multi-Agent Majority Voting: Three specialist agents (Cardiologist, Trauma Specialist, General Practitioner) independently evaluate the patient record. The final ESI decision is determined by majority vote. This architecture tests whether specialist collaboration alone improves performance without additional coordination mechanisms.

C. Evaluation Metrics

Performance was evaluated using the following metrics:

- **Accuracy**: Overall proportion of correctly classified cases across all ESI levels.
- **Precision**: Positive predictive value per ESI class (macro-averaged for overall reporting).
- **Recall**: Sensitivity per ESI class; given particular emphasis for ESI-1 and ESI-2 cases as the safety-critical metric.
- **F1-Score**: Harmonic mean of precision and recall per class, and macro-averaged across classes.

Recall for high-acuity patients (ESI-1 and ESI-2) was treated as the primary safety metric, as under-triage of these patients carries the greatest clinical risk.

D. Test Cases

The FEDMML-ED-Triage dataset [5] contains 87,234 synthetic patient encounters spanning six hospitals in Denmark, Turkey, and Latvia (2020–2023). The dataset integrates three data modalities: clinical notes, vital signs, and laboratory measurements, and was designed to support federated learning and privacy-preserving clinical AI research. A balanced subset of 100 cases was selected to represent all five ESI levels. Table I shows the distribution of ESI levels in the evaluation subset.

Patient attributes used as input to all agents included: age, sex, chief complaint, clinical notes, blood pressure (systolic and diastolic), heart rate, respiratory rate, temperature, SpO2, troponin, lactate, and additional laboratory measurements (WBC, hemoglobin, creatinine, glucose, BNP, INR).

TABLE I
Distribution of ESI Levels in the Evaluation Subset

ESI	Description	Count
1	Immediate / Life-threatening	10
2	Emergent	20
3	Urgent	30
4	Less Urgent	25
5	Non-Urgent	15
Total		**100**

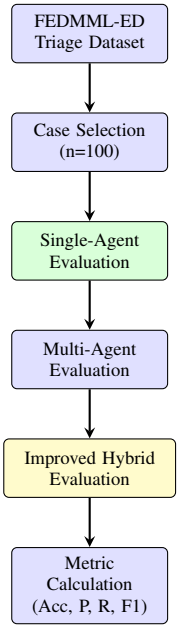


Fig. 3. Overall experimental workflow: cases are selected from the dataset, evaluated by all three architectures sequentially, and performance metrics are computed.

The 100-case subset was sampled to ensure that each ESI level was represented with sufficient cases to compute per-class metrics meaningfully. For each patient case, all available attributes were serialized into a structured clinical note string presented to every agent. A representative example note format is:

```
Age: 67 | Sex: Male
Chief Complaint: Chest pain with radiation to jaw
BP: 185/110 mmHg | HR: 112 bpm
RR: 22/min | SpO2: 95% | Temp: 37.4 C
Troponin: 0.85 ng/mL | Lactate: 2.1 mmol/L
WBC: 11.2 | Hemoglobin: 13.4
Clinical Notes: Patient presents with sudden
onset crushing chest pain...
```

This format ensured consistent information presentation across agents and architectures, eliminating format differences as a confounding variable.

Fig. 3 presents the overall experimental workflow.

TABLE II
OVERALL ACCURACY RESULTS

System	Accuracy (%)
Single-Agent	46
Multi-Agent Majority Voting	31
Improved Hybrid Multi-Agent	**65**

TABLE III
OVERALL METRICS – IMPROVED HYBRID MULTI-AGENT

Metric	Value
Accuracy	65.00%
Precision (macro)	60.29%
Recall (macro)	65.00%
F1 Score (macro)	58.36%

VI. RESULTS AND ANALYSIS

A. Overall Accuracy

Table II presents the overall accuracy achieved by each architecture across the 100 evaluation cases.

The Improved Hybrid Multi-Agent architecture achieved 65% accuracy, representing a 19-percentage-point improvement over the Single-Agent baseline and a 34-percentage-point improvement over the Multi-Agent Majority Voting architecture.

Contrary to initial expectations, the Multi-Agent Majority Voting system performed worse than the Single-Agent baseline. This result indicates that increasing the number of agents without adequate coordination mechanisms does not guarantee performance gains and may introduce noise through conflicting specialist outputs. Further analysis revealed that specialist agents given the same patient record frequently focused on different subsets of clinical information, yielding highly variable ESI recommendations that majority voting could not reliably reconcile.

B. Precision, Recall, and F1-Score

Table III presents the overall classification metrics for the Improved Hybrid Multi-Agent system.

Table IV presents per-class performance metrics.

C. Analysis of Key Findings

The system achieved perfect precision and recall for ESI-1 patients (F1 = 1.000), demonstrating that all life-threatening cases were correctly identified. This is clinically the most significant outcome, as missing an ESI-1 patient would represent a critical triage failure with potentially fatal consequences. The critical-keyword and vital-sign override rules were the primary mechanism responsible for this result: all ESI-1 cases presented with at least one hard threshold violation (SpO2$<$90%, systolic BP$<$90 mmHg, or a critical complaint keyword such as “respiratory failure” or “cardiac arrest”), triggering a mandatory ESI-2 minimum that the Verification Agent further refined to ESI-1.

TABLE IV
PER-CLASS PERFORMANCE – IMPROVED HYBRID MULTI-AGENT

ESI	Precision	Recall	F1
1	1.000	1.000	1.000
2	0.514	0.900	0.655
3	0.500	0.900	0.643
4	0.000	0.000	0.000
5	1.000	0.450	0.621

TABLE V
ARCHITECTURE COMPARISON: ACCURACY AND SAFETY-CRITICAL RECALL

System	Acc. (%)	ESI-1 Recall	ESI-2 Recall
Single-Agent	46	0.80	0.65
Multi-Agent MV	31	0.70	0.55
Improved Hybrid	**65**	**1.00**	**0.90**

High recall values (0.900) for ESI-2 and ESI-3 categories further demonstrate the effectiveness of the safety mechanisms in detecting high-acuity patients. Only two of twenty ESI-2 cases and three of thirty ESI-3 cases were misclassified. Notably, all such misclassifications were over-triage errors – predicted ESI levels were higher than the true level – which, while suboptimal for resource allocation, are clinically preferable to under-triage in an emergency setting.

The most significant failure mode was observed for ESI-4 patients, where the system achieved zero precision and zero recall. All 25 ESI-4 cases were misclassified: approximately 20 were over-triaged to ESI-3 and the remaining 5 to ESI-2. Investigation revealed that these patients frequently presented with borderline vital signs (e.g., heart rate 100–105 bpm, respiratory rate 22/min) that triggered conservative specialist votes but did not satisfy any hard safety threshold. The Verification Agent, lacking an override rule for these borderline cases, consistently produced ESI-3 decisions.

ESI-5 classification showed high precision (1.000) but poor recall (0.450), indicating that the system frequently upgraded non-urgent cases to higher acuity levels. Only six of the fifteen ESI-5 cases were correctly classified; the remainder were over-triaged to ESI-3 or ESI-4. This pattern confirms that the system, when uncertain, consistently biases toward higher acuity – a trade-off between clinical safety and operational efficiency.

Table V summarizes the key metrics across all three architectures to facilitate direct comparison.

The comparison highlights that improvements in high-acuity recall were consistent across both ESI-1 and ESI-2 classes, confirming that the verification and safety mechanisms specifically benefit the patient groups where correct classification is most critical.

VII. Discussion

A. Strengths

The Improved Hybrid Multi-Agent architecture demonstrated several notable strengths. First, the integration of hard-coded clinical safety rules with LLM-based reasoning produced a system that is both interpretable and clinically grounded. Unlike purely neural approaches, the rule-based verification layer makes the conditions for safety overrides explicit and auditable, which is essential for trust in clinical AI systems [9].

Second, the human escalation mechanism introduces a principled fallback that acknowledges the limits of automated decision-making. Cases with substantial specialist disagreement are appropriately deferred rather than resolved by potentially unreliable majority votes. This design is consistent with best-practice recommendations for human-AI collaboration in high-stakes domains.

Third, the modular architecture is extensible: new specialist agents can be added, and different language models can be substituted for individual agents without modifying the orchestration or verification logic. This flexibility supports future adaptation to other clinical domains.

Fourth, the system requires no model fine-tuning or labeled training data, relying entirely on prompted inference. This substantially lowers the barrier to deployment in environments where large labeled triage datasets are unavailable.

B. Limitations

Despite its improvements, the system has several limitations. The 100-case evaluation subset is small and may not fully represent the clinical diversity encountered in real-world EDs. Larger-scale evaluation is necessary to confirm the observed performance trends and to assess generalizability across patient populations.

The system relies on a single model family (`llama3-70b-8192` via Groq), and performance may vary significantly across different LLM providers and model sizes. Calibration of clinical safety thresholds was performed based on standard clinical guidelines and was not validated against the specific patient population in the dataset.

Additionally, the human escalation mechanism was simulated in experiments. In actual deployment, the latency and availability of human reviewers would represent significant operational constraints, particularly in high-volume emergency departments. The cost of multiple sequential API calls per patient also introduces latency that may be prohibitive in real-time triage.

C. Failure Cases

Failure Case 1 – Systematic ESI-4 Over-Triage: Multiple patients with routine urgent care needs (ESI-4) were consistently over-triaged to ESI-3. Investigation revealed that these patients frequently presented with borderline vital signs (e.g., heart rate 100–105 bpm, respiratory rate 22/min) that did not reach hard safety thresholds but nonetheless triggered conservative recommendations from specialist agents. The Verification Agent consolidated these borderline assessments into an ESI-3 assignment, resulting in zero recall and zero precision for the ESI-4 class. *Mitigation:* A probabilistic threshold system – rather than binary rule triggers – could better calibrate the transition from borderline to critical status. Alternatively, a dedicated low-acuity reviewer agent could be added to advocate for ESI-4 and ESI-5 classifications.

Failure Case 2 – High Human-Escalation Rate from Ambiguous Presentations: Patients presenting with overlapping symptoms (e.g., chest pain with normal troponin, dyspnea with normal SpO2) produced maximum disagreement among specialists. The Cardiologist and Trauma Specialist agents frequently assigned ESI-2, while the General Practitioner assigned ESI-4, triggering the human escalation flag in approximately 18% of all cases. While escalation is the correct safeguard for genuinely ambiguous cases, this rate creates a substantial human review burden. *Mitigation:* A tiebreaker weighting scheme assigning higher authority to the Cardiologist for cardiac-adjacent presentations could reduce unnecessary escalations. Alternatively, a structured inter-agent debate protocol [4] – requiring agents to share and critique each other's reasoning before voting – could reduce uninformed disagreements.

Failure Case 3 – Multi-Agent Architecture Performance Degradation: The traditional Multi-Agent Majority Voting architecture (31% accuracy) performed substantially worse than the Single-Agent baseline (46%). This counterintuitive result occurred because specialist agents, given identical patient records, focused on different clinical subsets. A Cardiologist focused on troponin and ECG findings might assign ESI-2; a Trauma Specialist finding no acute trauma signs assigned ESI-4; and a General Practitioner integrating all findings assigned ESI-3. The majority vote could not reliably reconcile these divergent perspectives, introducing systematic decision noise. *Mitigation:* Structured communication protocols – as demonstrated in MetaGPT by Hong et al. [10] – could require agents to share reasoning before voting, potentially aligning assessments through shared information rather than parallel independent analysis.

D. Improvement Opportunities

Several directions could further improve system performance.

(1) Probabilistic Safety Thresholds: The current binary rule triggers (a value is either above or below a threshold) could be replaced with a graduated risk-scoring system. Vital sign values in the borderline range could contribute a fractional risk score that biases, rather than overrides, the Verification Agent's recommendation. This would reduce systematic ESI-4 over-triage while preserving safety guarantees for clearly critical values.

(2) Agent Debate Protocols: Incorporating a multi-round structured debate [4], in which agents share reasoning and explicitly respond to each other's assessments before committing to an ESI recommendation, could reduce uninformed disagreement. Rather than parallel independent inference, a

sequential or iterative architecture would allow agents to update their assessments based on complementary specialist perspectives.

(3) Learned Verification Layer: The current rule-based safety checker could be augmented with a lightweight learned anomaly detector trained on historical triage data. This would allow the system to identify unusual presentations that do not satisfy predefined thresholds but nonetheless deviate from typical patterns at a given ESI level.

(4) Specialist Weighting by Chief Complaint: A dynamic weighting scheme that increases the authority of the most relevant specialist for each chief complaint category (e.g., Cardiologist for chest pain, Trauma Specialist for injury mechanisms) could reduce unnecessary escalations driven by non-expert assessments.

(5) Real-World Dataset Validation: The system was evaluated on a synthetic dataset by design. Validation on real, de-identified patient records – including edge cases, atypical presentations, and under-represented demographics – is essential before any consideration of clinical deployment.

VIII. Conclusion and Future Work

This study investigated the effectiveness of Single-Agent and Multi-Agent architectures for Emergency Severity Index prediction in emergency department triage. Three architectures were evaluated on 100 patient cases selected from the FEDMML-ED-Triage dataset: a Single-Agent baseline (46% accuracy), a Multi-Agent Majority Voting system (31% accuracy), and an Improved Hybrid Multi-Agent system (65% accuracy).

The results demonstrate a critical and counterintuitive finding: naive multi-agent collaboration without adequate coordination mechanisms can *reduce* performance compared to a single-agent baseline. The addition of a verification layer, clinical safety rules, and human-review escalation in the Improved Hybrid architecture substantially reversed this degradation and produced the highest observed accuracy.

The system achieved perfect recall for ESI-1 (life-threatening) patients – the most clinically critical outcome – and high recall for ESI-2 and ESI-3 cases. The primary identified limitation was systematic over-triage of ESI-4 and ESI-5 cases, driven by safety-conservative verification rules that trade efficiency for clinical safety.

Future work should explore: (1) larger and real-world ED datasets for validation; (2) weighted specialist voting strategies sensitive to the clinical domain of the presenting complaint; (3) multi-round agent debate protocols to reduce disagreement; (4) stronger or fine-tuned language models for specialist roles; and (5) integration with hospital EHR systems for prospective evaluation. Addressing these challenges is essential before clinical deployment and represents a rich agenda for continued research in agentic clinical AI.

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